

Nicholas Channg

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EDUCATION

Cornell University

Ithaca, NY

B.A. Computer Science, Minors in Artificial Intelligence, Data Science | GPA: 3.7/4.0 Expected Graduation, Dec 2026

- **Awards:** Y Combinator AI Startup School 2025 (1 of 2,500 selected), 4x Hackathon Winner

EXPERIENCE

Google

Sunnyvale, CA

Software Engineer Intern, Cloud Knowledge Catalog

May 2026 – Aug 2026

- Deployed low-cost, low-latency data profiling for 55M+ BigQuery External and BigLake tables across backend Java/Go microservices spanning 200K+ projects and 100K+ accounts, surfacing insights that ground Gemini Enterprise AI agents
- Developed dynamic query-generation logic based on table format, applying TABLESAMPLE or a LIMIT fallback with sampling rates computed from BigQuery row counts, reducing data scanned per profile by 90%+ vs. full-table scans
- Extended the user-facing API service to accept profiling requests for BigQuery external tables/views, adding type-aware validation and routing while preserving backward compatibility, rolled out behind Mendel feature flags
- Proactively expanded the Google Cloud SDK (gcloud) with a data profiling mode flag (Standard vs Lightweight), utilizing Python to modify API clients and build client-side validation to block incompatible customization inputs

Amazon

New York, NY

Software Engineer Intern, Amazon Ads

Sep 2025 – Dec 2025

- Developed and owned asynchronous Java and Python backend services in a distributed AWS video ads processing pipeline handling 80K+ jobs weekly, designing fault-tolerant workflows across SQS, ECS, Lambda, DynamoDB, and S3
- Built a production AI microservice on AWS Bedrock with a two-stage VLM-to-LLM pipeline to perform multimodal video analysis on video ads, generating titles/descriptions that outperformed advertiser-supplied text in 98% of cases
- Integrated a video assembly service into the pipeline by building a Lambda-triggered ECS execution path and S3/DynamoDB asset storage, enabling secure cross-account and cross-service access of final video assets

Tesla

Fremont, CA

Software Engineer Intern, Charging Team

May 2025 – Sep 2025

- Developed scalable REST API endpoints in the Tesla Mobile App backend with Node.js, Express.js, and TypeScript, orchestrating vehicle/charging microservice calls to power real-time user-visible data for 3.4M daily users
- Implemented a Redis caching layer to store Spark API responses for supercharger congestion/pricing data, reducing redundant upstream microservice calls and improving Tesla Mobile App response latency by 10%
- Built an agentic AI QA assistant with Python that automated mobile app test-account setup (e.g. region switching, vehicle linking) via orchestrated microservice API calls, improving QA throughput by 25%

Flow

Remote

Software Engineer Intern

Jan 2024 – May 2024

- Developed REST APIs using Flask and Python to retrieve and process data from Flow's AI models, automating customer prospecting and increasing sales representative call volume by 300% (30 → 120 calls/day)
- Reduced application latency by 15% by optimizing PostgreSQL queries; replaced nested SELECTs with JOINs, added B-tree indexes on high-cardinality columns, and batched execution to minimize database calls
- Integrated backend services into CI/CD pipelines using GitHub Actions, ensuring automated testing and deployment

PROJECTS

ElectAI | Python, Scikit-learn, Flask, React, TypeScript, TailwindCSS, NumPy, Pandas

Jan 2025

- Engineered a full-stack ML web app to predict voter turnout using historical behavioral data patterns to model participation likelihood using a Flask REST API to retrieve data from the model and a React frontend
- Built a data preprocessing pipeline with Python, NumPy, and Pandas to clean 20 years of election data, enabling accurate regression modeling with Scikit-learn and achieving high accuracy (RMSE: 5.35, MAE: 4.22)

TECHNICAL SKILLS

Languages: Java, Go, Python, C, TypeScript, JavaScript, SQL

Frameworks: Flask, FastAPI, Node.js, Express.js, React, Next.js

Cloud/Infrastructure: AWS, GCP, Redis, PostgreSQL, Docker, GitHub Actions, CI/CD

Libraries/Tools: Scikit-learn, NumPy, Pandas, OpenAI API, TailwindCSS, Git, Postman